Predicting S&P 500 Volatility Regimes: Milestone 1

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Code: github.com/mariyakoval/deep-learning (notebooks/milestone1.ipynb)

1. Task and data

We predict how volatile the S&P 500 (the SPY ETF) will be over the next 5 trading days: a 3-class regime (calm, middle, crisis) and the realized volatility itself. We predict volatility rather than direction because direction is close to random, while volatility is persistent and is what risk managers need.

Sources. Daily SPY prices and volume from yfinance; CPI, the 10Y-2Y Treasury spread and the fed funds rate from FRED. Financial news (FinBERT embeddings) is already collected but not used in this milestone. One row is one trading day: 4,773 days from 2006-01-06 to 2024-12-23.

Preprocessing. From prices we compute the log return, 21-day realized volatility, the daily high-low range, and volume relative to its trailing 1-year mean (raw volume trends over time). Macro series are shifted by their publication lag (CPI 45 days, fed funds 35, spread 1) so the model never sees a number before it was released. Features are standardized with train-split statistics.

Labels. There are no human labels. The target *fwd_rv* is the standard deviation of the next 5 daily log returns, annualized. The regime buckets *fwd_rv* at the 50th and 87th percentiles of the training split (10.6% and 21.3%). We first planned to label regimes with a hidden Markov model, but dropped it: those labels changed so rarely that "tomorrow equals today" was 97% accurate, and its decoding used future data.

date	log ret	rv 21	rel vol	hl range	fwd rv	regime
2020-01-02	0.0093	0.0706	-0.1034	0.0073	0.0966	calm
2020-03-16	-0.1159	0.7659	1.4734	0.0791	0.6724	crisis
2022-06-13	-0.0387	0.3170	0.7359	0.0225	0.3515	crisis

Table 1. Example rows (a calm day, the COVID crash, and the June 2022 sell-off).

Split. By date, never shuffled: train May 2009 to 2017, validation 2018-2019, test 2020-2023. The last 5 days of each split are dropped so no target overlaps the next split. A sample is the window of past days ending at the target day; the window may reach back before the split start, since past features are known at prediction time. The test set covers COVID, the 2022 bear market and the SVB collapse.

split	first	last	days	calm	middle	crisis
train	2009-05-01	2017-12-21	2,178	1086	808	284
val	2018-01-02	2019-12-23	498	281	145	72
test	2020-01-02	2023-12-21	1,001	292	453	256

Table 2. Split sizes and regime counts. The test set has far more crisis days (26%) than train (13%).

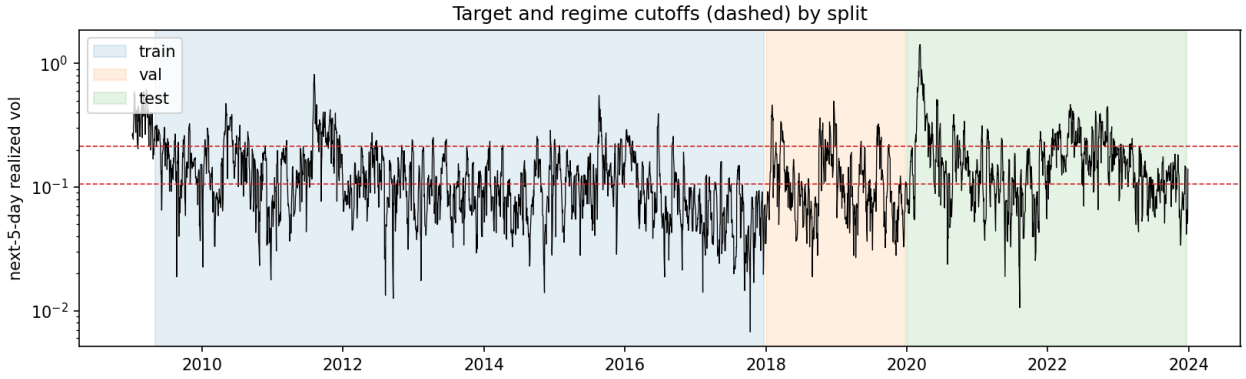


Figure 1. Target (log scale) with regime cutoffs (dashed) and the three splits.

2. Models

Baselines (0 parameters). *Majority*: always the most common train regime (calm) and the train median volatility. *Persistence*: the next 5 days look like the last 5, so predict the trailing 5-day realized volatility and the regime it falls in.

Simple model: 1D CNN (4,356 parameters). Input is the last 20 days of the 4 price features, shape (20, 4). Two Conv1d layers (16 channels, kernel 5, ReLU, max-pool 2), then flatten, dropout 0.2, a 32-unit linear layer with ReLU, and two output heads: 3 regime logits and one value for standardized log volatility. The loss is class-weighted cross-entropy (weights are inverse train class frequency, so missed crises cost more) plus MSE on log volatility. We use log volatility because raw volatility is very right-skewed.

3. Error metric and test results

Accuracy is misleading when one class is rare, and a missed crisis is worse than a false alarm. Our main metric is **macro-F1** over the three regimes, and we also report **crisis recall**. For volatility we report MAE and RMSE on the annualized scale. All models are scored on the same 1,001 test days; the CNN is the mean and standard deviation over 5 seeds.

model	params	macro-F1	crisis recall	accuracy	vol MAE	vol RMSE
majority	0	0.151	0.000	0.292	0.093	0.165
persistence	0	0.543	0.574	0.541	0.077	0.116
1D CNN	4,356	0.587 ± 0.015	0.765 ± 0.023	0.581 ± 0.015	0.069 ± 0.003	0.129 ± 0.009

Table 3. Test results, 2020-2023.

The CNN beats persistence on macro-F1 (0.587 vs 0.543), crisis recall (0.765 vs 0.574) and volatility MAE (0.069 vs 0.077). It loses on RMSE (0.129 vs 0.116): its forecasts are smoother, so it reacts late to sudden spikes such as March 2020, and RMSE punishes those few large errors heavily (Figure 2). Training on log volatility also pulls predictions toward typical levels. During the 2022 bear market the CNN caught 94% of crisis days versus 65% for persistence.

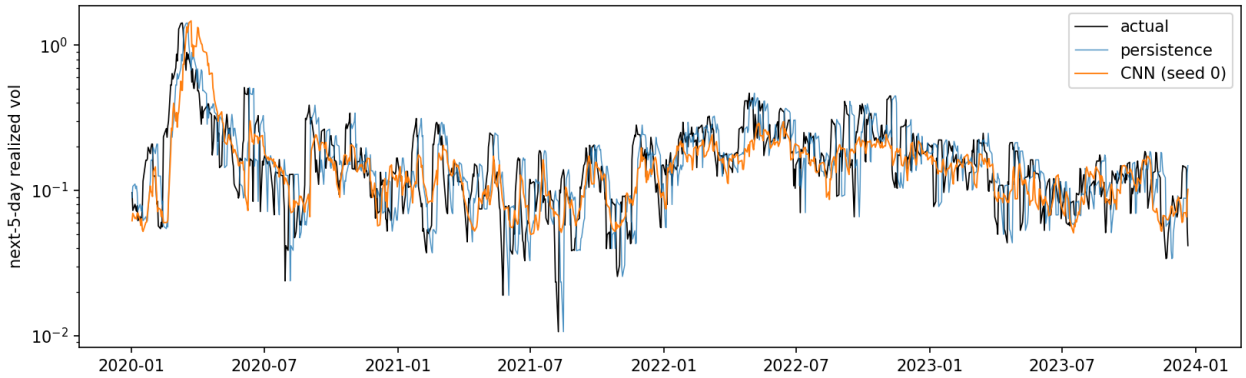


Figure 2. Test-period volatility: actual, persistence, and the CNN (seed 0).

4. Optimisation

Adam (weight decay 1e-4), batch size 64, up to 100 epochs, early stopping on validation loss with patience 10, keeping the best weights. We ran a grid on the validation set only: feature set (price + macro, price, price with relative volume) x window (20, 60) x channels (16, 32) x learning rate (1e-3, 3e-4), 3 seeds each, 72 runs. We picked the setting with the lowest mean validation loss: price with relative volume, window 20, 16 channels, learning rate 0.001 (validation loss 1.363 ± 0.046).

Two choices mattered most. **Macro features hurt badly** (best validation loss 2.12 vs 1.36): interest rates and inflation in 2018-2023 were far outside the training range. Fed funds in the test years reaches 19 training standard deviations above its training mean, so the network gets inputs it has never seen. **A 20-day window beat 60 days** for every feature set: with about 2,200 training samples, the longer window gives a larger linear layer (9-21k vs

4-11k parameters) that overfits sooner. Figure 3 shows validation loss flattening after 7-15 epochs while training loss keeps falling, which is where early stopping keeps the model.

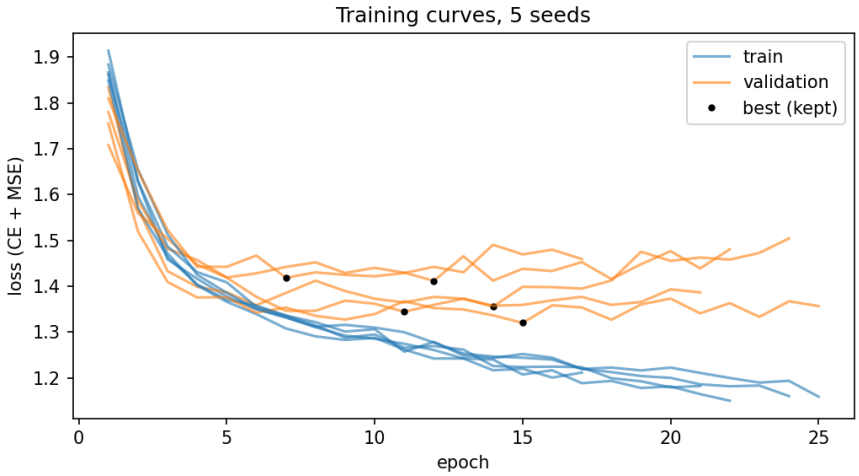


Figure 3. Training and validation loss for the selected model, 5 seeds. Dots mark the kept epoch.

Reproducibility. The notebook runs top to bottom on CPU in about 5 minutes from a committed snapshot of the processed table. Seeds are fixed, PyTorch runs in deterministic mode and package versions are pinned; two clean runs gave identical results files.

5. What next

Over the next three weeks we will add the news branch to answer our main question: feed the daily FinBERT embeddings (available for 98.5% of trading days in 2009-2023) into the same network and compare it with this price-only CNN on the same test days. We will also try macro inputs as changes rather than levels so they do not fall outside the training range, and try a loss on the raw volatility scale to reduce the large errors at spikes.